

Modular Relative Jacobian for Dual-Arms and the Wrench Transformation Matrix

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Abstract—A modular relative Jacobian is recently derived and is expressed in terms of the individual Jacobians of stand-alone manipulators. It includes a wrench transformation matrix, which was not shown in earlier expressions. This paper is an experimental extension of that recent work, which showed that at higher angular end-effector velocities the contribution of the wrench transformation matrix cannot be ignored. In this work, we investigate the dual-arm force control performance, without necessarily driving the end-effectors at higher angular velocities. We compare experimental results for two cases: modular relative Jacobian with and without the wrench transformation matrix. The experimental setup is a dual-arm system consisting of two KUKA LWR robots. Two experimental tasks are used: relative end-effector motion and coordinated independent tasks, where a force controller is implemented in both tasks. Furthermore, we show in an experimental design that the use of a relative Jacobian affords less accurate task specifications for a highly complicated task requirement for both end-effectors of the dual-arm. Experimental results on the force control performance are compared and analyzed.

I. INTRODUCTION

A relative Jacobian is used to control the dual-arm as a single manipulator (with one end-effector). This approach drastically increases the dimension of the combined null space, as compared to controlling each individual manipulators separately. For example, when each manipulator of a dual-arm has seven degrees-of-freedom (7-DOF) and are controlled in full Cartesian space individually, the combined null space has only 2-DOF. However, through the use of a relative Jacobian, the relative motion between the two end-effectors is considered such that at full Cartesian space control between the two end-effectors, an 8-DOF combined null space results.

The use of relative Jacobian [1], [2] has recently gained renewed attention because of the increased studies in dual-arms control [3]–[9], to name a few. Two major advantages can be enumerated: (1) all the principles of single manipulator control can be applied to dual-arm manipulators, and (2) a dramatic increase in the null-space dimension. The first formulations of the relative Jacobian [1], [2] analytically

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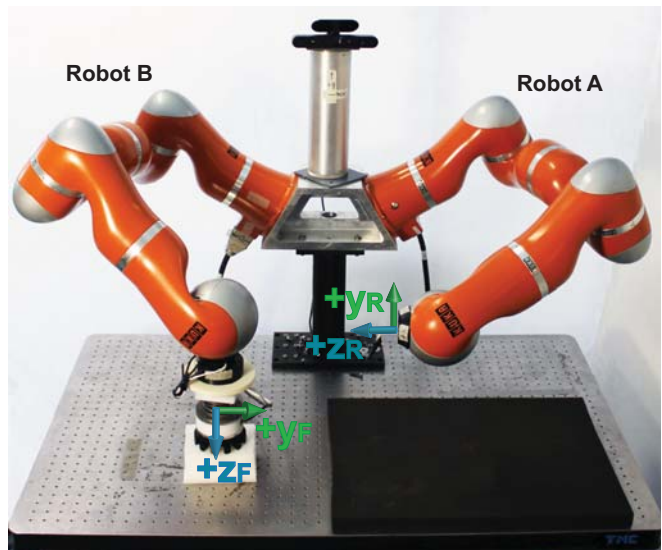


Fig. 1. The KUKA dual-arm manipulator setup. The right arm (robot B) end-effector holds the tool, while the left arm (robot A) end-effector is the reference end-effector. Axes z_R and y_R are the relative position axes located at the robot A end-effector frame, while z_F and y_F are the force sensor axes located at the force sensor frame. The tool end-effector exerts a normal force on the table (along z_F), while the two end-effectors move relative to each other (horizontally along z_R).

computed it based on the assigned kinematic chain between the two end-effectors. In order to simplify its implementation, one approach is to express the relative Jacobian in modular form, that is, in terms of the Jacobian of each stand-alone manipulator.

Just recently, a newly derived modular relative Jacobian [10] reveals a wrench transformation matrix, which was not shown in previous expressions [11]–[13] or were in an unsimplified, non-compact form [14]–[16]. This new and recent expression drastically simplifies the relative Jacobian implementation because one only needs to consider the individual Jacobians for each manipulator, and then arrives at the relative Jacobian of the combined manipulators by performing corresponding Jacobian transformations as shown in [10]. In the same work, it was analytically proven that the exclusion of the wrench transformation matrix may result in an end-effector motion, for a null space motion when the wrench transformation is included. Furthermore, it was experimentally shown that at higher end-effector velocities, the effect of the wrench transformation matrix cannot be ignored.

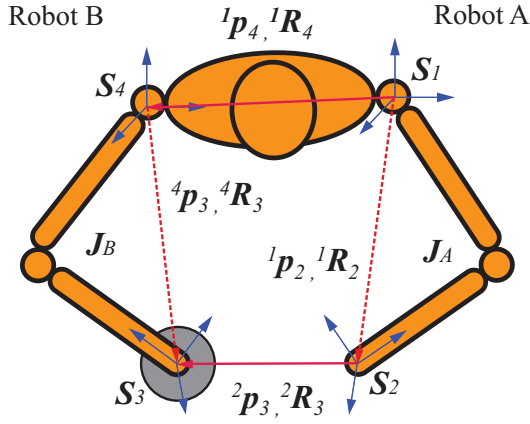


Fig. 2. Schematic diagram of the frame assignments for the dual-arm in Fig. 1. The tool end-effector (frame S_3) is expressed with respect to the reference end-effector (frame S_2). The corresponding position and orientation transformations are shown in the diagram. The Jacobian for robot A is denoted J_A and the Jacobian for robot B is denoted J_B .

This work extends the experimental investigation in [10] by evaluating a dual-arm force controller, without necessary driving the end-effectors at higher velocities. We investigate two cases: a modular relative Jacobian with (newly derived) and without (previously derived) the wrench transformation matrix. We perform two experimental tasks: relative end-effector motion and coordinated independent tasks. Both force and motion controllers [17]–[19] are implemented in full relative operational space for both experimental task executions. More recent studies on cooperative manipulation include trajectory planning for manipulation of flexible structure [20], manipulation and transportation of aerial robots [21], [22], decomposition frameworks with unilateral thrusters [23], and cooperative dual-arm mobile manipulation [24].

II. THEORETICAL BACKGROUND

To express the modular relative Jacobian J_R of a dual-arm, we let the Jacobian of the left arm (reference end-effector) to be J_A , and the Jacobian of the right arm (tool end-effector) to be J_B , as shown in Fig. 2. From the same figure, the position ${}^i\mathbf{p}_j$ and rotation ${}^i\mathbf{R}_j$ are expressed with respect to the indicated reference frames. Thus the modular relative Jacobian as derived in [10] can be expressed as

$$\mathbf{J}_R = [-{}^2\Psi_3 {}^2\Omega_1 \mathbf{J}_A \quad {}^2\Omega_4 \mathbf{J}_B] \quad (1)$$

where

$${}^2\Psi_3 = \begin{bmatrix} \mathbf{I} & -S({}^2\mathbf{p}_3) \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \quad \text{and} \quad {}^i\Omega_j = \begin{bmatrix} {}^i\mathbf{R}_j & \mathbf{0} \\ \mathbf{0} & {}^i\mathbf{R}_j \end{bmatrix}. \quad (2)$$

The matrix ${}^2\Psi_3$ is the wrench transformation matrix expressed in the indicated reference frames, while the matrix ${}^i\Omega_j$ is a rotation matrix in the corresponding reference frames. The symbol $S(\mathbf{p})$ is a skew-symmetric matrix with

input vector $\mathbf{p}$, that is,

$$S(\mathbf{p}) = \begin{bmatrix} 0 & -p_z & p_y \\ p_z & 0 & -p_x \\ -p_y & p_x & 0 \end{bmatrix}. \quad (3)$$

Previous expressions [11]–[13] of the modular relative Jacobian $\mathbf{J}'_R$ assumes the wrench transformation matrix ${}^2\Psi_3$ to be an identity matrix. In this work, we compare the dual-arm force and motion control performance in using $\mathbf{J}_R$ and $\mathbf{J}'_R$.

For mobile manipulators, the relative Jacobian $\mathbf{J}_{RT}$ was derived in [10] as,

$$\mathbf{J}_{RT} = [-{}^2\Psi_3 {}^2\Omega_0 \mathbf{J}_{bA} \quad -{}^2\Psi_3 {}^2\Omega_1 \mathbf{J}_A \quad {}^2\Psi_3 {}^2\Omega_0 \mathbf{J}_{bB} \quad {}^2\Omega_4 \mathbf{J}_B] \quad (4)$$

where $\mathbf{J}_{bA}$ is the Jacobian of the mobile base A and $\mathbf{J}_{bB}$ is the Jacobian of the mobile base B. The frame assignment is the same as in Fig. 2 but the frames S_1 and S_4 are now moving.

III. IMPLEMENTATION

To implement the proposed relative Jacobian to the KUKA dual-arm, we define the corresponding terms in (1). We denote the position and orientation of the robot A end-effector with respect to its own base as $\mathbf{p}_A, \mathbf{R}_A$; and the robot B end-effector position and orientation with respect to its own base as $\mathbf{p}_B, \mathbf{R}_B$. Then the following assignments are true: ${}^1\mathbf{R}_2 = \mathbf{R}_A$, ${}^1\mathbf{p}_2 = \mathbf{p}_A$, ${}^4\mathbf{R}_3 = \mathbf{R}_B$, and ${}^4\mathbf{p}_3 = \mathbf{p}_B$.

From (2), we can get

$${}^2\Omega_3 = \begin{bmatrix} \mathbf{R}_A^T & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_A^T \end{bmatrix} \quad \text{and} \quad {}^2\Omega_4 = \begin{bmatrix} \mathbf{R}_A^T {}^1\mathbf{R}_4 & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_A^T {}^1\mathbf{R}_4 \end{bmatrix} \quad (5)$$

and from [10], we can express the relative position between the end-effectors ${}^2\mathbf{p}_3$ as

$${}^2\mathbf{p}_3 = \mathbf{R}_A^T {}^1\mathbf{p}_4 + \mathbf{R}_A^T {}^1\mathbf{R}_4 \mathbf{p}_B - \mathbf{R}_A^T \mathbf{p}_A \quad (6)$$

where the reference frames S_1 and S_4 (function of ${}^1\mathbf{p}_4, {}^1\mathbf{R}_4$ and are normally static) are dependent on how robots A and B are mounted on the shoulders of the torso.

A. Torso Mounting

The shoulder mounting of our torso has a 60-degree inclination from the horizontal and another 30-degree rotation around the normal to the inclined plane. From this shoulder configuration, the orientation ${}^1\mathbf{R}_4$ and position ${}^1\mathbf{p}_4$ are defined as

$${}^1\mathbf{R}_4 = \mathbf{R}_{z,-30} \mathbf{R}_{y,-120} \mathbf{R}_{z,150} \quad (7)$$

and

$${}^1\mathbf{p}_4 = \mathbf{R}_{z,-30} \mathbf{R}_{y,-150} \mathbf{p}_{bases} \quad (8)$$

where $\mathbf{p}_{bases} = [0, 0, 0.22]^T$ m. The symbol $\mathbf{R}_{i,a}$ denotes rotation of a degrees around the i -th axis.

B. Gravity Compensation

Due to the change in the configuration of the base mounting, the KUKA robot need to be corrected in terms of the its gravity compensation vector $\mathbf{g} = [0, 0, 9.81]^T$. The modified gravity compensation for the right arm is

$$\mathbf{g}_{right} = \mathbf{R}_{z,30} \mathbf{R}_{y,-60} \mathbf{g} \quad (9)$$

and the left arm is

$$\mathbf{g}_{left} = \mathbf{R}_{z,-30} \mathbf{R}_{y,-60} \mathbf{g}. \quad (10)$$

This results into $\mathbf{g}_{right} = [-7.36, -4.25, 4.91]^T$ and $\mathbf{g}_{left} = [-7.36, 4.25, 4.91]^T m/s^2$.

C. Torque Controller

We use the KUKA Controller 30, defined as each-axis controller. To implement a force and motion controller on the KUKA, its impedance controller is given as

$$\tau_{cmd} = \mathbf{J}^T (\mathbf{k}_c (\mathbf{x}_{FRI} - \mathbf{x}_{msr}) + \mathbf{F}_{FRI}) + D(\mathbf{d}_c) + f_{dynamics}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}) \quad (11)$$

where τ_{cmd} is the commanded torque to the robot joints, $\mathbf{J}^T$ is the manipulator Jacobian, $\mathbf{k}_c$ is the motion control proportional gain, $\mathbf{x}_{FRI}$ is the desired end-effector position and orientation, $\mathbf{x}_{msr}$ is the measured end-effector position and orientation, $\mathbf{F}_{FRI}$ is the force/torque input, $\mathbf{d}_c$ is the motion damping gain, and $f_{dynamics}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$ is the robot dynamics model. From the variables in (11), the fast research interface (FRI) of KUKA (which is its programming interface), only allows modification of $\mathbf{k}_c$, $\mathbf{x}_{FRI}$, $\mathbf{F}_{FRI}$, and $\mathbf{d}_c$.

To overwrite the KUKA controller, we set $\mathbf{x}_{FRI} = \mathbf{x}_{msr}$ (or $\mathbf{k}_c = \mathbf{0}$). The force and motion controller is then assigned to $\mathbf{F}_{FRI}$, and the values of $\mathbf{d}_c$ are set at the default values. From (11), the resulting KUKA controller then becomes

$$\tau = \mathbf{J}^T \mathbf{F}_{FRI} \quad (12)$$

such that (11) becomes

$$\tau_{cmd} = \tau + D(\mathbf{d}_c) + f_{dynamics}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}). \quad (13)$$

Thus for each robot A and B manipulators, the corresponding KUKA controller is set to $\tau_A = \mathbf{J}_A^T \mathbf{F}_{FRI,A}$ and $\tau_B = \mathbf{J}_B^T \mathbf{F}_{FRI,B}$. These terms are then added to their corresponding damping and dynamics terms to get the commanded torque for each manipulator. To implement the dual-arm controller, the controller torques for both robots are set at,

$$\begin{bmatrix} \tau_A \\ \tau_B \end{bmatrix} = \mathbf{J}_R^T \mathbf{F}_R + (\mathbf{I} - \mathbf{J}_R^T \mathbf{J}_R^{T+}) [\mathbf{J}_A \ \mathbf{0}]^T \mathbf{F}_N \quad (14)$$

where $\mathbf{F}_R$ is the controller for both motion and force in the relative operational space, and $\mathbf{F}_N$ is the motion controller in the null space.

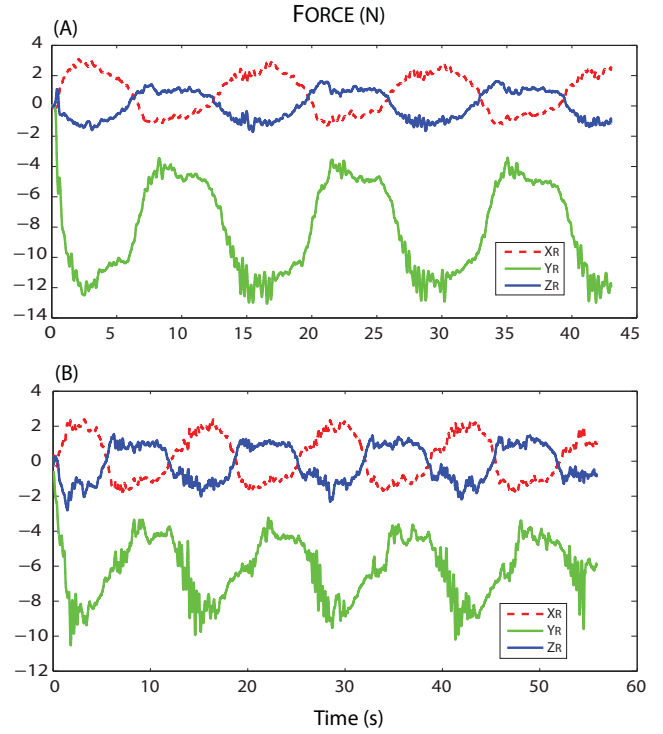


Fig. 3. Force sensor feedback expressed at the reference end-effector frame. The normal force exerted by the tool end-effector is along the $\mathbf{y}_R$ -axis, while both end-effectors are moving horizontally along the table in a relative motion with respect to each other. Case $\mathbf{J}_R$ is shown in subfigure A, while case $\mathbf{J}_R'$ is shown in subfigure B.

IV. EXPERIMENT I: RELATIVE END-EFFECTOR MOTION

The KUKA dual-arm platform posed two major challenges to our force and motion control experimental results. First, it does not give access to any low-level real-time controller. And second, to simultaneously control the two robots, we can only achieve a maximum sampling frequency 125 Hz (8 ms). Because of these limitations, the force and motion controller gains were not optimally tuned. However given these constraints, the performed experiments still showed comparative results (although non-optimal) between the use of $\mathbf{J}_R$ and $\mathbf{J}_R'$ that substantiate the effect of the wrench transformation matrix in the force control performance. A hybrid force/motion controller is implemented all throughout the experiments.

In the first experiment, we refer to Fig. 1 for the controller setup. In the relative operational space, the two end-effectors will move with respect to each other along the table (along $\mathbf{z}_R$ -axis), while the tool end-effector will exert a normal force on the table (along $\mathbf{z}_F$ -axis). To achieve this, force is controlled along the relative $\mathbf{y}_R$ -axis, with a desired force of -20 N. A desired sinusoidal motion along the relative $\mathbf{z}_R$ -axis is specified as

$$z_{R,des} = A(\cos(\omega t) - \cos(\omega t - \phi)) \quad (15)$$

where $A = 0.5$ m is the desired amplitude, $\omega = 0.15\pi$ rad/s is the desired angular frequency, t is current time, and $\phi = 10$ degrees is the desired phase shift, which determines

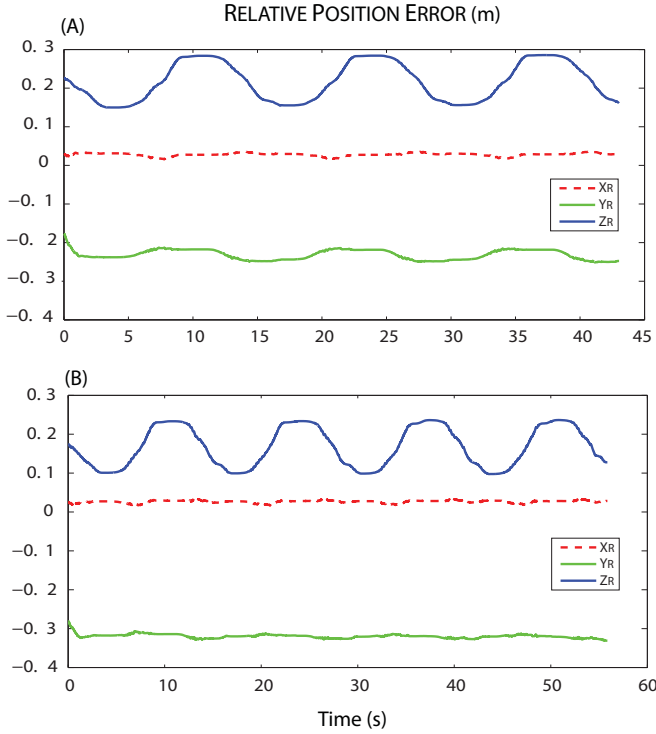


Fig. 4. Relative position error of tool end-effector motion with respect to the reference end-effector: case $\mathbf{J}_R$ is shown in subfigure A and case $\mathbf{J}'_R$ is shown in subfigure B.

the magnitude of the incremental step size. The remaining relative axes are in motion control, specified to maintain the initial position and orientation. Proportional position gains are set at 200 while proportional orientation gains are at 100. Proportional force gain is set at 0.2.

Fig. 3 showed the force control performance between $\mathbf{J}_R$ (subfigure A) and $\mathbf{J}'_R$ (subfigure B). The $\mathbf{x}_R$ - and $\mathbf{y}_R$ -axes are controlled in motion and are showing force feedback along those directions to be approximately the same. With a desired force along $\mathbf{y}_R$ set to -20 N , $\mathbf{J}_R$ exerted an average range of around $[-4, -12]\text{ N}$, while $\mathbf{J}'_R$ only exerted an average range of around $[-4, -10]\text{ N}$. A maximum normal force of -2 N difference with $\mathbf{J}_R$ case being closer to the desired force. Thus even with a non-optimal controller, the effect of the wrench transformation matrix showed better performance in force control. We believe that at a higher sampling frequency (around 1 kHz , that is eight times faster than the current setup), the force control performance comparison can be better.

Relative position error between the tool and reference end-effectors are shown in Fig. 4 for $\mathbf{J}_R$ (subfigure A) and $\mathbf{J}'_R$ (subfigure B). Along the $\mathbf{x}_R$ -axis, the desired position is the initial position just before the task execution. In both $\mathbf{J}_R$ and $\mathbf{J}'_R$ cases, the error is approximately the same. Along the $\mathbf{y}_R$ -axis, position is not controlled because this is the force control direction. However, the error value with respect to the initial position can be indicative on how well the force controller is performing, with the $\mathbf{J}_R$ case having 0.1 m less error than $\mathbf{J}'_R$. Along the $\mathbf{z}_R$ -axis, the average error of the $\mathbf{J}_R$

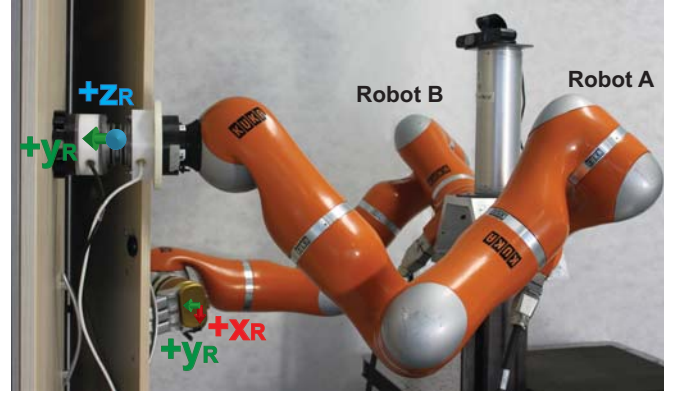


Fig. 5. The KUKA dual-arm manipulator performing a coordinated independent task: robot B opening and closing a cabinet door by moving along the $\mathbf{y}_R$, and robot A exerting a normal force (along $\mathbf{y}_R$) on the other cabinet door while moving along an oscillatory motion along $\mathbf{z}_R$.

case is around 0.22 m , while for the $\mathbf{J}'_R$ case it is around 0.18 m , a difference of around 0.04 m , with $\mathbf{J}_R$ case being slightly higher. This slightly higher position error in the $\mathbf{J}_R$ case can be attributed to the fact that the controller gains were not optimally set and are therefore much more sensitive to noise and disturbances. The higher normal force exertion along the $\mathbf{y}_R$ -axis in the $\mathbf{J}_R$ case created higher friction that resists the motion of the tool end-effector and subsequently, a bigger disturbance to the motion control in the $\mathbf{z}_R$ direction, resulting into a slightly higher relative position error along $\mathbf{z}_R$.

V. EXPERIMENT II: COORDINATED INDEPENDENT TASKS

The second experiment is for robot B to open and close a cabinet door, while robot A exerts a normal force on the other cabinet door and moving in an oscillatory manner. This experiment is designed to show two things: (1) the ease of implementing motion for the reference end-effector, and (2) the capability of the relative Jacobian to show seemingly independent tasks, but were performed through relative motion (or force).

A. Experimental Design

We note that robot B (reference/gripper) end-effector is in the null-space of the dual-arm manipulator, thus its absolute location can be arbitrary. This makes the opening and closing of the door task much easier to implement. Because the reference end-effector grips the cabinet door handle (and its absolute location is arbitrary), a human can open and close the cabinet door and the reference end-effector will be compliant to this motion, that is, it will keep gripping the cabinet door and the rest of the robot arm will move to accommodate such an end-effector displacement.

The ease of implementation has two aspects. First, we only need to specify the oscillatory displacement along the $\mathbf{y}_R$ -axis for the gripper end-effector, without specifying any desired displacement along the $\mathbf{z}_R$ -axis to accommodate the arc movement of the door handle in the operational space. Second, given a specified amplitude of oscillation, the start

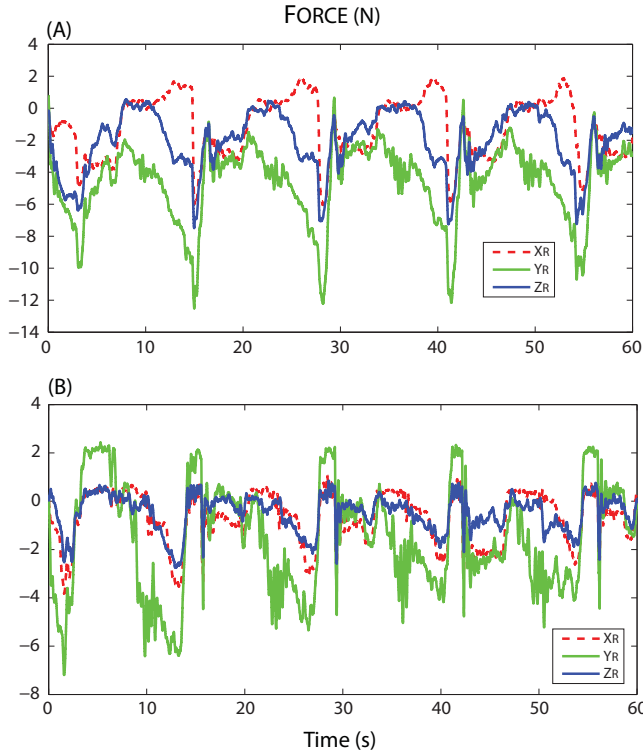


Fig. 6. Force sensor feedback expressed with respect to the reference end-effector frame for the coordinated independent tasks experiment. The normal force exerted on the other cabinet door is along y_R . Case J_R is shown in subfigure A and case J'_R is shown in subfigure B.

location of the oscillatory motion can be arbitrary because once the gripper hits the maximum allowable displacement (cannot push further against the cabinet) along the y_R -axis, the tool end-effector moves to adjust the relative motion between the two end-effectors. However, the tool end-effector is force controlled (no motion control) along the y_R -axis, so it will not move but will keep on exerting a normal force on the other cabinet door.

The second point of the experimental design is the independence of the task execution despite using relative motion between the end-effectors. The tool end-effector is force controlled along the y_R -axis, while moving in an oscillatory motion along the z_R -axis (compared to the oscillatory movement of the gripper end-effector along y_R). Thus both tool and gripper end-effector will appear oscillating along two independent axes, while the tool end-effector exerts a normal force on the other cabinet door. Although the tool end-effector's oscillating motion along z_R needs to adjust to the small displacement of the gripper in this direction (due to the arc movement of the door handle), this motion is not critical to the tool end-effector's task execution.

B. Experimental Results

While the gripper end-effector is opening and closing one cabinet door, the tool end-effector is oscillating on the surface of the other cabinet door and exerting a normal force on the surface. The normal force feedback is shown in Fig. 6. Again, because KUKA does not allow access to a low-level,

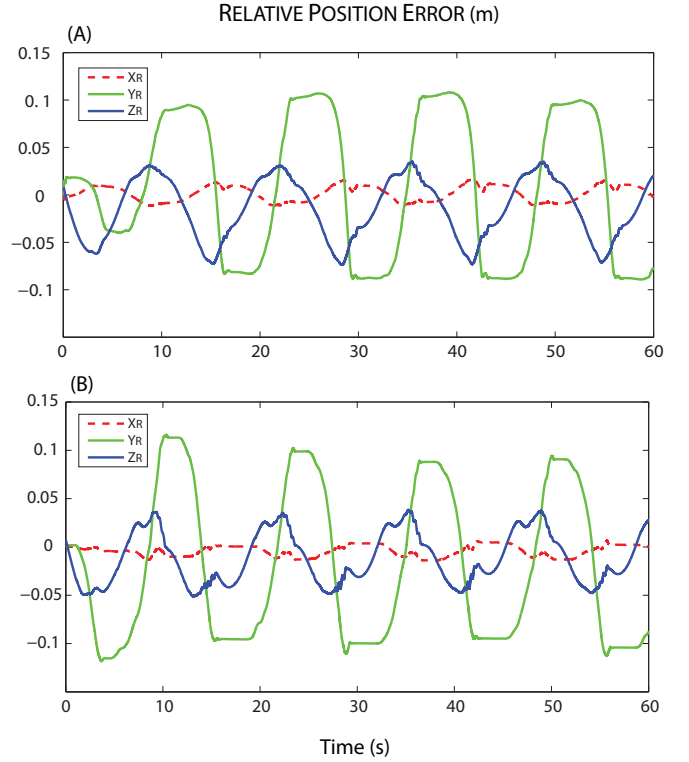


Fig. 7. Relative position errors between tool end-effector and reference end-effector during coordinated independent tasks experiment. The J_R case is shown in subfigure A, while J'_R case is shown in subfigure B.

real-time controller and the sampling frequency is at 125 Hz, the force control is expected to be non-optimal. To keep the tool end-effector to maintain contact during task execution, we specify a desired normal force of -30 N . For case J_R , it is able to exert a normal force in the average range of around $[0, -11]\text{ N}$ (subfigure A), while case J'_R is within the average range of around $[2, -5]\text{ N}$ (subfigure B) and losing surface contact. As seen in these results, with a much more complicated task execution between the end-effectors, a much more superior force control response was shown by J_R , exerting around double the maximum force compared to J'_R . At the same time, the latter case consistently loses contact as indicated by an offset non-contact force of around 2 N . (This offset force value was set just before task execution, and enhances visualization of the non-contact case. This implies that the actual normal force exerted is slightly higher the force reading. It is noted that the offset is present in both cases, but not necessarily equal in value.) And because J'_R case consistently loses contact during the oscillatory motion, it has lesser value of the range of force along the x_R and z_R axes compared to the J_R case.

The relative position errors during the coordinated independent tasks are shown in Fig. 3, with case J_R in subfigure A and case J'_R in subfigure B. The y_R -axis is force controlled and is therefore not controlled in position, such that in both cases the position errors along this axis are comparable. The x_R axis is fixed (distance from the floor) during task execution, and so the relative position errors along this axis

is comparable for both cases. Lastly, the error range along the $\mathbf{z}_R$ -axis is slightly higher in $\mathbf{J}_R$ than $\mathbf{J}'_R$. Considering the non-optimality of the controller settings, there is a greater disturbance in $\mathbf{J}_R$ because of the greater frictional force during oscillatory motion. This is true because of the bigger exertion of the normal force, as well as being consistently in contact with the surface of the door cabinet, as compared to the $\mathbf{J}'_R$ case that kept on losing contact with the door surface.

VI. CONCLUSION

This paper has shown the contribution of the wrench transformation matrix to the force control using modular relative Jacobian. It clearly indicated its influence by resulting into a more accurate force control performance even at low end-effector velocities. Two experiments were performed to show the difference in performance with and without the wrench transformation matrix. In both experiments, the tool end-effector was required to exert a normal force on a surface, while both tool and reference/gripper end-effectors moved with respect to each other. Experimental results showed that the case with the wrench transformation matrix included has superior force control performance, with a difference of 2 N and 5 N in maximum force exerted in each respective experiment. As the task became much more complicated, the superiority of the force control performance with the wrench transformation matrix became much more evident, doubling its maximum exerted force compared to the case without this matrix. Furthermore, this paper also showed the ease of implementing two independent tasks, through the use of relative Jacobian: opening and closing a door on one end-effector, and exerting a normal force with oscillatory motion on the other end-effector. This complicated task execution could have required more accurate task specifications if these tasks were implemented independently with dedicated controllers for each end-effector. Lastly, we believe that the experimental difference will be much more pronounced given better controller setup, that is, with access to low-level, real-time controller and at a much higher sampling frequency.

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